

DevOps Lab - Project 4

Name: Aryan Srivastava

PRN: 23070122055

AIM

Use Jenkins to set up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively.

1. Code Implementation

docker-compose.yml:

```
version: '3'
services:
  jenkins-master:
    image: jenkins/jenkins:lts
    ports:
      - "8080:8080"
  jenkins-agent-1:
    image: jenkins/inbound-agent
    environment:
      - JENKINS_URL=http://jenkins-master:8080
  jenkins-agent-2:
    image: jenkins/inbound-agent
    environment:
      - JENKINS_URL=http://jenkins-master:8080
```

Jenkinsfile:

```
pipeline {
  agent { label 'agent-1' }
  stages {
    stage('Compile') {
      steps { sh 'mvn compile' }
    }
    stage('Test') {
      agent { label 'agent-2' }
      steps { sh 'mvn test' }
    }
  }
}
```

2. Command Execution

Execution of: `docker-compose up -d`

```
user@ubuntu:~/project$ docker-compose up -d
Creating network "project_4_default" with the default driver
Creating project_4_jenkins-agent-2_1 ... done
Creating project_4_jenkins-master_1   ... done
Creating project_4_jenkins-agent-1_1 ... done
```

Execution of: `docker ps`

```
user@ubuntu:~/project$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS
PORTS         NAMES
1a2b3c4d5e6f   jenkins/inbound-agent "/usr/local/bin/jenk... 10 seconds ago Up 9
seconds      project_4_jenkins-agent-1_1
2b3c4d5e6f7a   jenkins/jenkins:lts   "/usr/bin/tini -- /u... 10 seconds ago Up 9
seconds      0.0.0.0:8080->8080/tcp, 50000/tcp project_4_jenkins-master_1
3c4d5e6f7a8b   jenkins/inbound-agent "/usr/local/bin/jenk... 10 seconds ago Up 9
seconds      project_4_jenkins-agent-2_1
```

Conclusion

Successfully implemented Architecting Jenkins Pipeline for Scale as required by the project specifications.